Polynomial Slater preparations and uniform orbit densities

Exact neutral-slope pinching on the selected polynomial core

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Abstract

We extend the finite-N linear preparation estimate to fixed Gaussian degree in the half-filled double-scaled regime. Partial traces of the coefficient seed have weights within a controlled distance of a Slater weight. A Weyl-orbit count bounds the dimensions of every supporting irreducible representation. It controls all oriented tensor flattenings of the orbit synthesis map. Iterated fixed-Schatten Gaussian inequalities and a conditional Wick identity then give dimension-uniform density moments. This transfers exact energy-pinching errors to every fixed selected polynomial preparation. Two quadratic preparations give explicit normalized slope correlators. Multiple interlaced nonpolynomial pinched factors are not inferred from convergence on this core.

1 Setting and quantitative statement

Use standard CAR and the same overlapping Hermitian complex Gaussian p-body couplings as in the preceding Slater drafts. Set $m = N/2$, $d_m = \binom{N}{m}$, $\mathcal{H}_m = \bigwedge^m \mathbb{C}^N$ and fix a Slater determinant O. Real independent standard Gaussian coordinates g_a carry the orthogonal noise representation $R(U)$ of $U(N)$. Let $P_O(g)$ have degree at most $K \geq 1$ and obey

$$P_O(k \cdot g) = k_m P_O(g) k_m^\dagger, \qquad k \in U(m) \times U(N-m). \tag{1}$$

Write $P_O O = \sum_{k=0}^K P_{k,O} O$ for its orthogonal Wick-chaos decomposition and $a_k = \mathbb{E}\,\|P_{k,O}O\|^2$, $a = \sum_k a_k$. All these quantities may depend on N.

Theorem 1. *Assume $p^2/N \leq \Lambda$ and $p \leq N/(12K)$, and put*

$$C_K = e^{36K^2\Lambda+1}, \qquad B_{0,r} = 1, \qquad B_{k,r} = \frac{[k(2r-1)]^k}{k!} C_K^{1-1/r} \quad (k \geq 1). \tag{2}$$

There is a positive covariant orbit density R_P such that for covariant Y in joint Schatten L^2,

$$\mathbb{E}\langle O|P_O^\dagger Y P_O|O\rangle = \mathbb{E}\,\mathrm{tr}_m(R_P Y), \tag{3}$$

$$\|R_P\|_{L^r(\mathrm{tr}_m)} \leq \left(\sum_{k=0}^K \sqrt{B_{k,r} a_k}\right)^2 \leq (K+1) \max_{k \leq K} B_{k,r}\, a. \tag{4}$$

Here $r \geq 1$ is an integer and $\mathrm{tr}_m = \mathrm{Tr}\,/d_m$.

The constants are not optimized. For fixed K and $p^2/N \to \lambda > 0$, the hypotheses hold eventually. This is a fixed-degree statement, not a bound uniform in degree or a general matrix-polynomial hypercontractivity assertion.

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2 Orbit synthesis and the coefficient seed

Normalized Haar measure is understood. Define

$$q_U(g) = U_m^\dagger P_O(U \cdot g)O, \qquad Z(g)f = \sqrt{d_m}\int q_U(g)f(U)\,\mathrm{d}U, \qquad R_P = ZZ^\dagger. \tag{5}$$

Only a finite-dimensional space of matrix coefficients is needed in the Haar domain. Gaussian invariance and $Y(U \cdot g) = U_m Y(g) U_m^\dagger$ give (3). They also give $\mathbb{E}\,\mathrm{tr}_m R_P = a$. Chaos projection commutes with orthogonal noise rotations, so (1) holds separately in each chaos.

For degree k, use a symmetric coefficient tensor in ordered indices:

$$P_{k,O}(g)O = \sum_{i_1,\dots,i_k} A_{i_1\cdots i_k} : g_{i_1}\cdots g_{i_k} :, \qquad \Psi = \sum_{\boldsymbol{i}} e_{i_1}\otimes\cdots\otimes e_{i_k}\otimes A_{\boldsymbol{i}}. \tag{6}$$

Wick isometry in this ordered symmetric convention reads

$$a_k = k!\,\|\Psi\|^2\,. \tag{7}$$

Let $\rho = R^{\otimes k}\otimes\bigwedge^m$. The coefficient tensor of q_U is $\Psi_U = \rho(U^{-1})\Psi$. For a diagonal unitary D, stabilizer covariance and $D_m O = \chi_O(D)O$ imply

$$\rho(D)\Psi = \chi_O(D)\Psi, \qquad w = (\underbrace{1,\dots,1}_{m},0,\dots,0). \tag{8}$$

This is a statement about the full tensor, before taking any partial trace.

Complexify the real Hermitian noise space and choose an orthonormal weight basis. It is unitarily equivalent to the conjugate adjoint representation on $\mathrm{End}(\bigwedge^p\mathbb{C}^N)$. Its weights are differences of two p-subset indicators. Every noise weight ν has integral entries, sum zero and $\|\nu\|_1 \le 2p$. This change of coefficient basis does not change any flattening's singular values; we still use real independent Gaussians for the probabilistic estimates.

Trace out a set J of ℓ noise legs. In a product weight basis,

$$T_I = \mathrm{Tr}_{\mathcal{W}^{\otimes J}}\,|\Psi\rangle\langle\Psi| = \sum_\beta |\psi_\beta\rangle\langle\psi_\beta|, \quad I = [k]\setminus J. \tag{9}$$

Each nonzero ψ_β has a weight $z = w - \nu_\beta$ with

$$\sum_i z_i = m, \qquad \|z - w\|_1 \le 2\ell p. \tag{10}$$

Thus partial tracing loses the exact Slater weight but retains a quantitative neighborhood of it. This is the additional input needed beyond the preceding linear argument.

3 A near-Slater Weyl-orbit bound

Lemma 1. *Under the hypotheses of Theorem 1, every integral weight satisfying (10), with $0 \le \ell \le K$, has at least d_m/C_K distinct coordinate permutations. Every irreducible $U(N)$ representation containing such a weight therefore has dimension at least d_m/C_K.*

Proof. For $\ell = 0$ the orbit has exactly d_m elements. Otherwise let b be the number of entries of z outside $\{0, 1\}$ and let S be their sum. Since $z - w$ is integral of total sum zero, its total positive and negative masses are both at most ℓp. Consequently

$$b \le 2\ell p, \quad -\ell p \le S \le b + \ell p, \quad |S|, |S - b| \le 3\ell p =: D. \tag{11}$$

The counts of ones and zeros are $n_1 = m - S$, $n_0 = m - b + S$. Put $\delta_1 = -S$, $\delta_0 = S - b$; then $|\delta_i| \le D \le m/2$ and $\delta_0 + \delta_1 = -b$.

For an integer $|\delta| \le m/2$,

$$\log \frac{(m+\delta)!}{m!} \le \delta \log m + \frac{\delta^2 + |\delta|}{m}. \tag{12}$$

For positive δ, expand the logarithm as $\delta \log m + \sum_{j=1}^{\delta} \log(1 + j/m)$ and use $\log(1 + x) \le x$. For $\delta = -t$, expand it as $-t \log m - \sum_{j=0}^{t-1} \log(1 - j/m)$ and use $-\log(1 - x) \le 2x$ for $0 \le x \le 1/2$. Both give (12), including the endpoints.

If n_t counts the entries equal to $t \notin \{0, 1\}$, the orbit cardinality $L(z)$ satisfies

$$\frac{L(z)}{d_m} = \frac{m!^2}{n_0! n_1! \prod_{t \ne 0,1} n_t!} \tag{13}$$

$$\ge \frac{m^b}{b!} \exp\left[-\frac{2D^2 + 2D}{m}\right] \ge \exp[-36\ell^2\Lambda - 1] \ge C_K^{-1}. \tag{14}$$

We used $\prod n_t! \le b!$, $b \le m$, and $2D/m = 12\ell p/N \le 1$. Permutation matrices implement the Weyl group. Their distinct weights have orthogonal weight spaces in a unitary representation, so its dimension bounds their number from below. □

No multiplicity-free assumption is involved. The estimate permits entries such as 2 and -1 as well as imbalanced numbers of zeroes and ones. It is deliberately coarse; the bounded p^2/N and fixed K assumptions, rather than closeness in operator norm, keep its loss independent of N.

4 All oriented flattenings and decoupled moments

Let F_I flatten the synthesis coefficient tensor with physical output and noise legs I on its row side, and the Haar input and remaining noise legs J on its column side. Summing over the column noise indices cancels their unitary rotations. Thus

$$F_I F_I^\dagger = d_m \int \rho_I(U^{-1}) T_I \rho_I(U^{-1})^\dagger \, \mathrm{d}U, \qquad \|F_I\|_{\mathrm{HS}}^2 = d_m \, \|\Psi\|^2 \, . \tag{15}$$

Decompose the row representation as $\bigoplus_\alpha V_\alpha \otimes \mathcal{M}_\alpha$. Every supporting irreducible component of T_I contains one of the weights in (10): project each ψ_β in (9) onto the isotypic components, using that these projections commute with the torus. Schur averaging gives

$$F_I F_I^\dagger = \bigoplus_\alpha \frac{d_m}{\dim V_\alpha} I_{V_\alpha} \otimes \mathrm{Tr}_{V_\alpha}(T_{I,\alpha\alpha}) \le C_K \, \|\Psi\|^2 \, I. \tag{16}$$

Here each multiplicity matrix is positive with norm no larger than its trace, and $\mathrm{Tr}\, T_I = \|\Psi\|^2$. Hence, for all orientations,

$$\|F_I\|_{S_{2r}}^{2r} \le d_m C_K^{r-1} \, \|\Psi\|^{2r} \, . \tag{17}$$

For clarity we derive the exact probabilistic form required here. For $X = \sum_i g_i A_i$, Hermitian dilation and the fixed-Schatten Gaussian inequality give

$$\mathbb{E}\, \|X\|_{S_{2r}}^{2r} \le \frac{(2r-1)^r}{2} \left[\mathrm{Tr} \left(\sum_i A_i A_i^\dagger \right)^r + \mathrm{Tr} \left(\sum_i A_i^\dagger A_i \right)^r \right] . \tag{18}$$

Indeed, Gaussian integration by parts bounds $\mathbb{E}\,\mathrm{Tr}\, S^{2r}$ by $(2r-1)\mathbb{E}\,\mathrm{Tr}(V|S|^{2r-2})$. Trace Hölder followed by scalar Hölder gives $(\mathbb{E}\,\mathrm{Tr}\, S^{2r})^{1/r} \le (2r-1)(\mathrm{Tr}\, V^r)^{1/r}$; dilation contributes the factor $1/2$. This is the standard matrix-Gaussian argument [2].

Let Z_{dec} use independent Gaussian families for its k slots. Condition on $k-1$ families and apply (18). The two terms are the $2r$-moments of a row concatenation and a column stacking. Repeat in each branch. Each noise slot is assigned once to a row or column, producing precisely all 2^k oriented flattenings:

$$\begin{aligned} \mathbb{E}\, \|Z_{\mathrm{dec}}\|_{S_{2r}}^{2r} &\le (2r-1)^{kr} 2^{-k} \sum_{I \subset [k]} \|F_I\|_{S_{2r}}^{2r} \\ &\le d_m (2r-1)^{kr} C_K^{r-1} (a_k/k!)^r . \end{aligned} \tag{19}$$

The flattening iteration is established methodology [1]; we retain fixed Schatten moments, not a logarithmic-dimensional operator-norm specialization.

5 Coupled Wick chaos and sums of degrees

Let $g^{(1)}, \dots, g^{(k)}$ be independent standard Gaussian vectors and let $g = k^{-1/2} \sum_j g^{(j)}$. Conditional means are $g/\sqrt{k}$; conditional cross-covariances between distinct slots are $-I/k$. The conditional exponential generating function is

$$\mathbb{E}\left[\exp\left(\sum_j t_j \cdot g^{(j)}\right)\middle| g\right] = \exp\left[\frac{g \cdot \sum_j t_j}{\sqrt{k}} + \frac{1}{2}\sum_j |t_j|^2 - \frac{1}{2k}\left|\sum_j t_j\right|^2\right]. \tag{20}$$

Differentiate once in each slot. Terms involving two derivatives of the same t_j cannot contribute. Comparing with the Wick generating function yields, including repeated coordinate indices,

$$\mathbb{E}\left[\prod_{j=1}^{k} g_{i_j}^{(j)}\middle| g\right] = k^{-k/2} : g_{i_1} \cdots g_{i_k} : . \tag{21}$$

Consequently $Z_k(g) = k^{k/2}\mathbb{E}[Z_{\mathrm{dec}} \mid g]$ exactly. Convexity of the $2r$-th power of a Schatten norm gives

$$\mathbb{E}\,\|Z_k\|_{S_{2r}}^{2r} \le k^{kr}\,\mathbb{E}\,\|Z_{\mathrm{dec}}\|_{S_{2r}}^{2r}. \tag{22}$$

With (7) and (19), this proves

$$\left(d_m^{-1}\mathbb{E}\,\mathrm{Tr}(Z_k Z_k^\dagger)^r\right)^{1/r} \le B_{k,r}a_k. \tag{23}$$

This proof handles Wick polynomials with repeated indices directly; no square-free-chaos restriction is imported from an operator-norm decoupling theorem.

The zero-chaos vector has weight w in $\bigwedge^m \mathbb{C}^N$, whose weight space is one-dimensional, so $P_{0,O}O = cO$. Its orbit density is $|c|^2 I$ and $B_{0,r} = 1$ is exact. Since $Z = \sum_k Z_k$, Minkowski in the Gaussian–Schatten space, normalized by d_m, gives

$$\|R_P\|_{L^r}^{1/2} = d_m^{-1/(2r)}\,\|Z\|_{L^{2r}(S_{2r})} \le \sum_{k=0}^{K} \|R_{P_k}\|_{L^r}^{1/2}. \tag{24}$$

Cauchy–Schwarz and Gaussian-chaos orthogonality finish (4). This argument retains interference between degrees; it does not assert $R_P = \sum_k R_{P_k}$ samplewise.

For $k = 1$, the preceding linear-stage estimate $\mathbb{E}\,\mathrm{tr}_m R_{P_1}^r \le (2r-1)!!a_1^r$ is stronger and holds without half filling or double scaling. The present constants are only used to secure finite, N-independent bounds at every fixed degree.

6 Exact pinching on the polynomial core

For a covariant X_N, (3) and Hölder imply

$$\mathbb{E}\,\|X_N P_N O\|^2 \le \|R_{P,N}\|_{L^2(\mathrm{tr}_m)}\,\|X_N\|^2_{L^4(\mathrm{tr}_m)}\,. \tag{25}$$

Take P_N to be the evaluation of a fixed polynomial in the previously constructed $h_N, u_N, M_N, M_N^\dagger$. Its Gaussian degree is fixed, it obeys stabilizer covariance, and its total a_N stays bounded by the established same-J fixed-word limits. Thus any covariant error going to zero in tracial L^4 goes to zero as a vector in every such preparation.

In particular, use the actual normalized unpinched slope from the neutral-slope stage,

$$t_N = \frac{m\epsilon\alpha(\Phi_m(H_N) - \bar\kappa_m H_N)}{8\lambda_N C_m^{3/2} q_N \sqrt{\lambda_N}}, \qquad b_N = (1 - \mathcal{P}_{H_N})t_N, \tag{26}$$

where $h_N = H_N/\sqrt{C_m}$, $\lambda_N = p^2/N$, $q_N = e^{-4\lambda_N}$, $\alpha = 1 - p/m$, $\epsilon = 1 - \alpha^2$. The symbols $\Phi_m, \bar\kappa_m$ retain their exact covariance-channel and regression definitions from that stage; $\mathcal{P}_{H_N}$ keeps entire degenerate energy blocks. Its tracial results give

$$\|\mathcal{P}_{H_N} t_N\|_{L^r(\mathrm{tr}_m)} \longrightarrow 0, \qquad \|t_N - u_N\|_{L^r(\mathrm{tr}_m)} \longrightarrow 0 \quad \text{for every fixed finite } r. \tag{27}$$

The second statement also follows by interpolating the rank/size L^2 bridge with the uniform higher moments. Equation (25) therefore yields

$$\mathbb{E}\,\|(b_N - u_N)P_N O\|^2 \longrightarrow 0. \tag{28}$$

These are dimensionless string-energy observables with $\hbar = 1$, fixed $\lambda > 0$ and annealed half-filled Slater expectations. No thermal or gravitational interpretation is added by this estimate.

More generally, let W_N be a fixed covariant block of h_N, u_N, b_N and fixed-time h_N phases. Tracial Hölder and telescoping show $W_N(b_N) - W_N(u_N) \to 0$ in L^2; fixed higher norms bound all other factors. Apply (3) and polarize P_N, Q_N to obtain

$$\varphi_N(P_N^\dagger W_N(b_N) Q_N) \longrightarrow \varphi(P^\dagger W(h,u) Q). \tag{29}$$

The unpinched limit on the right is the established Slater transfer representation, not the tracial vacuum unless its preparation says so. An arbitrary number of selected factors is allowed in the fixed polynomials P, Q; the exact-pinched factors remain inside a covariant block. This does not yet justify raw words with several such blocks separated by selected operators: their intervening preparations are no longer Gaussian polynomials. Nor does core convergence alone establish those high moments.

7 Quadratic preparations and explicit slope responses

In the subset labels below, O and E denote the occupied and unoccupied one-particle index sets. Then

$$M_N^\dagger = L_N^{-1/2} \sum_{A\subset E,\, B\subset O} J_{AB} C_A^\dagger C_B, \quad L_N = \binom{m}{p}\binom{N-m}{p}, \tag{30}$$

with $|A| = |B| = p$ and GUE covariance $\mathbb{E}J_{AB}J_{CD} = \delta_{AD}\delta_{BC}$. The normalization of h_N removes the common physical coupling scale. For $P_N = M_N^\dagger h_N$, a possible internal contraction pairs the left selected label (A, B) with the right label (B, A). The latter annihilates particles in $A \subset E$ and hence kills O. For $P_N = (M_N^\dagger)^2$, a selected label's reverse is never selected. Both preparation vectors consequently have zero chaos-zero part and are pure second Wick chaos. This argument is finite N.

The established transfer rule, with $\gamma = q^{1/4}$, $q = e^{-4\lambda}$ and $s = \sqrt{q}$, gives

$$a_{M^\dagger h} = \varphi(hMM^\dagger h) = q(1+s), \tag{31}$$

$$a_{(M^\dagger)^2} = \varphi(M^2(M^\dagger)^2) = 2q. \tag{32}$$

Both are positive at fixed $\lambda > 0$. At small finite N, the second vector can vanish when $2p > m$; no normalization is taken there. Let the normalized preparation expectation divide by its finite-N a_N. Applying (29) to b_N^2 and evaluating the existing same-J word rule yields

$$\lim_N \frac{\varphi_N(h_N M_N b_N^2 M_N^\dagger h_N)}{\varphi_N(h_N M_N M_N^\dagger h_N)} = 1 + \frac{\lambda s q(1+q)}{1+s}, \tag{33}$$

$$\lim_N \frac{\varphi_N(M_N^2 b_N^2 (M_N^\dagger)^2)}{\varphi_N(M_N^2 (M_N^\dagger)^2)} = 1 + 2\lambda s q(1+q). \tag{34}$$

For example, before normalization the first numerator is $q[1 + s + \lambda\gamma^6(1+q)]$, and the second is $2q[1 + 2\lambda\gamma^6(1+q)]$. The code checks these word polynomials in algebraically independent γ and $\sqrt{\lambda}$ before imposing the physical curve.

The exact finite fixtures retain the same disorder and CAR signs:

N	p	$\mathbb{E}\left\|M_N^\dagger h_N O\right\|^2$	$\mathbb{E}\left\|(M_N^\dagger)^2 O\right\|^2$
4	1	2/3	1/2
6	1	11/12	8/9
4	2	1/6	0
6	2	1/6	0
8	2	2/9	1/18

Here $m = N/2$. These are finite algebra checks, not fits to the double-scaled asymptotic values in (32).

8 Evidence, attribution and continuation

The exact driver checks the original finite CAR preparations, symmetrizes their ordered Gaussian coefficient tensors and compares the Wick factor 2! with a separate pairing of raw coefficients. The saved output includes:

- 1,610 full seed-weight identities and 6,440 partial-weight checks over five finite half-filled CAR systems and the two preparations;
- 20,199 rational factorial estimates and 342 near-weight cases, including asymmetric entries outside $\{0, 1\}$;
- 126 conditional Wick identities through degree six, with repeated indices;
- eight exact moment comparisons for rectangular tensors of degrees one through three, checking all flattening orientations and the coupled-chaos Jensen bound;
- symbolic transfer expressions for the two preparation norms and their two unnormalized u^2 responses.

These diagnostics do not replace the representation-theoretic and probabilistic proofs above. They use no Haar Monte Carlo or floating point extrapolation. The scientific JSON records parameters, software versions and input SHA256 hashes. A separate read-only manifest binds this draft, its PDF, source notes, build logs and parent evidence. The successful driver is not rerun or overwritten during PDF production.

The use of tensor flattenings, Gaussian matrix inequalities and decoupling is established methodology. Our project-specific step is the control of every synthesis flattening by the near-Slater weight estimate in the half-filled double-scaled regime. Equation (21) is derived here to fix the repeated-index and normalization conventions, rather than proposed as a new general decoupling principle. Existing q-word rules supply the algebra in (33)–(34); the new input is the finite-N transfer of exact pinching into these preparation states.

The next microscopic interface is the quadratic covariant generator J_2 with selected M in the same disorder, followed by the full bilocal in this representation. The previously completed full-trace bilocal four-point construction is retained. Interlaced pinched moments, physical bulk spatial and mass generators and finite-G dynamics remain open; no full de Sitter dictionary is claimed here.

References

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[2] J. A. Tropp, *Second-Order Matrix Concentration Inequalities*, §§6–7, especially the proof of Proposition 7.1. https://tropp.caltech.edu/papers/Tro16-Second-Order-Matrix-preprint.pdf.

[3] A. Day and Chengheng, *Uniform Slater-orbit densities for linear selected preparations*, local project draft, 8 November 2026; and the preceding *Slater neutral-slope* and *Slater covariant* stages. Manuscripts, scientific outputs and provenance in this repository.